

[48]

SARDAR PATEL UNIVERSITY
Master of Information Technology M Sc IT Semester II Examinations
PS02CINT53 – Artificial Intelligence



Time: 10:00 a.m. to 1:00 p.m.

Date: 12/04/2025

Marks: 70

Q1. Choose the most appropriate option for each question.

[8]

- 1 The aim of AI is to impart _____.
(A) Generic intelligence (B) Intelligence in narrow domain
(C) Automation (D) None of these
- 2 _____ falls under the expert tasks category of AI.
(A) Financial analysis (B) Perception (C) Translation (D) None of these
- 3 In fuzzy set theory, the degree of membership of an element is always _____.
(A) 0 (B) 1 (C) Between 0 and 1 (D) None of these
- 4 Fuzzy logic can be used to handle _____.
(A) Uncertainty (B) Self-learning (C) Evolution (D) None of these
- 5 Bio-inspired computing takes inspiration from _____ and solves problems more effectively and intelligently.
(A) Swarms (B) Nature (C) Birds and Animals (D) All of these
- 6 Decisions like “to join a job or not” can be taken by a _____.
(A) Perceptron (B) Multi layer ANN (C) Hopfield network (D) None of these
- 7 If a function Y is defined as $Y=f(x)=3*x+7$. What is the fitness of an individual if $x=0$?
(A) 0 (B) 7 (C) 10 (D) None of these
- 8 GA is used when _____.
(A) Solution space is large (B) Traditional solutions not available
(C) Both A and B (D) None of these

Q2. Answer the following questions (Any Seven):

[14]

- 1 List the constituents of artificial intelligence.
- 2 What is KBS shell?
- 3 List advantages of KBS.
- 4 Define Union operation of two fuzzy relations.
- 5 List any two fuzzy rules of your choice in any domain.
- 6 What does an artificial neuron consist of? Give an example.
- 7 List the three models/architectures of neural network.
- 8 List two application areas where GA should be used.
- 9 Define schema in GA. Also, give an example.

[P.T.O.]

Q3. Answer the following questions:

- A Explain various types of knowledge with examples in detail. [6]
B Explain (i) forward chaining and (ii) backward chaining in an inference engine. [6]

OR

- B Explain the production system in detail by taking a suitable example. [6]

Q4. Answer the following questions:

- A Explain the concept of fuzzy membership functions. Develop fuzzy membership functions for a two-wheeler's High, Moderate, and Low speed. [6]
B Consider $U = V = \{1,2,3\}$. The relationship R , of $U \times V$, defined as "approximately equal," is a binary fuzzy relationship given by $1/(1,1)$, $1/(2,2)$, $1/(3,3)$, $0.8/(1,2)$, $0.8/(2,3)$, $0.8/(2,1)$, $0.4/(1,3)$, $0.4/(3,1)$, and $0.4/(3,2)$. Consider another relationship, S , of $U \times V$ defined as "x considerably larger than y" for all x in U and all y in V given by $0/(1,1)$, $0/(2,2)$, $0/(3,3)$, $0.6/(1,2)$, $0.6/(2,3)$, $0/(2,1)$, $0.8/(1,3)$, $0/(3,1)$, and $0/(3,2)$. Show :
(i) The relationship R and S in proper matrix format.
(ii) Find and show $R \cup S$ (Union) in proper matrix format.
(iii) Find and show $R \cap S$ (Intersection) in proper matrix format.

OR

- B Explain fuzzy rule based system by showing its general architecture. [6]

Q5. Answer the following questions:

- A Explain the working of a single perceptron by taking an example of your choice. [6]
B Design a neural network to select a mobile by taking suitable parameters of your choice. [6]

OR

- B Design a neural network to recommend a meal plan based on the following example training data. [6]

Sr. No.	X1 Calories	X2 Protein	X3 Carbs	X4 Age	X5 Gender	X6 Disease	X7 Life Style	O1 Plan A	O2 Plan B	O3 Plan C
1	0.8	0.6	0.4	0.4	0	0	0.5	1	0	0
2	0.5	0.4	0.7	0.5	0	1	0.6	0	1	0
3	0.5	0.4	0.2	0.2	1	0	0.4	0	0	1

Q6. Answer the following questions:

- A Explain the use of GA by giving its outline and diagram for function optimization by taking a suitable example of your choice. [6]
B Maximize $y = f(x) = 2 \cdot x$, where all x belong to $[0,7]$ using genetic algorithm. What is the value of x that maximizes the function Y? What is the maximum value of the function Y? [6]

OR

- B Consider a typical traveling salesperson problem with two routes R1 and R2 as follows.
 $R1 = \{1,3,2,4,5,6\}$
 $R2 = \{5,4,1,3,2,6\}$
Generate (i) an adjacency matrix and (ii) at least two new routes called R3 and R4 for the given problem.

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